

Introduction to Control Systems (ME2801)

Homework #8: Frequency Response

1. Ratio → Decibels

The magnitude (AKA gain) of the transfer function is often expressed in decibels (dB) in the frequency response. The following table illustrates expressing the magnitude portion of the frequency response as both a Ratio of Input/Output and in Decibels ($20 \cdot \log_{10}(\text{Ratio})$).

Fill in the missing values in the table. Ideally, you can complete this without using a calculator.

Magnitude of Transfer Function G	
Ratio [In/Out]	Decibels [dB]
1	0
10	
	40
1000	
1E+06	120
1E+09	
	-20
	-40
0.001	-60
1E-06	
1E-09	
2	6
4	
8	
16	
	-6
	-12
	-18
	-24

Magnitude of Transfer Function G	
Ratio [In/Out]	Decibels [dB]
1	0
10	20
100	40
1000	60
1E+06	120
1E+09	180
0.1	-20
0.01	-40
0.001	-60
1E-06	-120
1E-09	-180
2	6
4	12
8	18
16	24
0.5	-6
0.25	-12
0.125	-18
0.0625	-24

2. Reading Magnitude Graph

Given the sinusoidal inputs below, find the steady-state output amplitude with appropriate units.

1. V_{in} : Amplitude of 0.1 V at 0.01 rad/s.
2. V_{in} : Amplitude of 1.0 V at 0.1 rad/s.
3. V_{in} : Amplitude of 0.1 V at 1 rad/s.

